

Checkpoint

14.3: Partial Derivatives

Problem 1. *Why is the derivative ambiguous in 3-dimensions?*

Problem 2. *What else do we need in order for a derivative to no longer be ambiguous? What are the most natural choices for these?*

Problem 3. *Find f_x and f_y where $f(x, y) = 3xy + x^3 - 7y^2$.*

Problem 4. *What is $f_{xxxx...x}$ if $f(x, y) = ye^x$*

Problem 5. *Find $\frac{\partial f}{\partial y}$ where $f(x, y) = \ln(xy + x)$.*

1 Answers

1. There are infinitely many lines tangent to a point on a surface.
2. We need to specify a direction. $\hat{i}$ and $\hat{j}$ are the most natural choices (x -direction and y -direction).
3. $f_x = 3y + 3x^2$ and $f_y = 3x - 14y$.
4. ye^x
5. $\frac{x}{xy+x}$